area is a protein [18]. The results of solid-state nuclear magnetic resonance analysis indicate that the low molecular mass compounds and glycoproteins in aggregate glue droplets play key roles in their adhesive properties [19], that is, the loss of the low molecular mass compounds leads to protein hardening [19,20]. Therefore, a suspended aggregate glue droplet means that the protein is in a dissolved state, while an attached aggregate glue droplet means that the protein is precipitated. In the "fried egg" structure of the attached aggregate glue droplet, low molecular mass compounds and proteins should be separated and complementary in distribution. Choline in low molecular mass compounds is a common ionic liquid cation, and dihydrogen phosphate is a common ionic liquid anion. This suggests that the low molecular mass compounds may be ionic liquids. Here, we assume that the low molecular mass compounds of aggregate glue droplets are ionic liquids that dissolve proteins. If the assumption holds, the choline cation in low molecular mass compounds as the counterion of dihydrogen phosphate should exhibit the same distribution as Na^+, K^+, and dihydrogen phosphate. In this study, we characterized the inorganic and organic components of aggregate glue droplets at atomic and molecular levels using in situ measurement methods of the time-of-flight secondary ion mass spectrometry (TOF-SIMS), mid-infrared hyperspectral imaging, fluorescence microscopy, second harmonic generation microscopy, and scanning electron microscope energy dispersive X-ray spectrometry. The composition distribution observed by the mapping images of mass spectrometry and infrared absorption spectroscopy confirmed the above hypothesis and provided experimental evidence for the attachment mechanism of aggregate glue droplets.

Results

1. Fluorescence of aggregate glue droplets.

Figures 1a and 1d are the optical microscopic images of suspended and attached aggregate glue droplets with white light illumination respectively. Figures 1b and 1e are the fluorescence microscopy images of

suspended and attached aggregate glue droplets excited by an ultraviolet light source respectively and observed by a CCD camera with a long pass filter with a cut-on wavelength at 420 nm. Figures 1c and 1f are the optical spectra of fluorescence of suspended and attached aggregate glue droplets by a fiber spectrometer, respectively. In Fig. 1d, the opaque granules at the core of each droplet are apparent. The opaque granules are hardened glycoprotein which acts as an anchor to the silk [21]. The optically transparent region around the granule is considered that is a glycoprotein glue [21]. On the other hand, in the fluorescence microscopy image of attached aggregate glue droplets of Fig. 1e, the fluorescence signals were observed from the granules and the glycoprotein glue region. The fluorescence signals from the granules are stronger than that from the glycoprotein glue region. Interestingly, in Fig. 1e, the outer layer of the glycoprotein glue region shows a non-luminous structure. It suggests that this area has no fluorescence activity. While in Fig. 1d, the area corresponding to this outer layer is optically transparent in the direction perpendicular to the substrate and cannot be optically distinguished from the glass substrate. The results of subsequent mass spectrometry mapping confirmed that the non-luminous area in Fig. 1e is the exudate of low molecular mass compounds and inorganic salts. Therefore, the origin of the fluorescent signal in Fig. 1e was determined to be the glycoprotein, namely, the glycoprotein of aggregate glue droplet is a fluorescent protein.

In the suspended aggregate glue droplets, fluorescence is emitted from the whole of the aggregate glue droplet, namely the distribution of glycoprotein in suspended aggregate glue droplets is uniform. Moreover, the fluorescence spectra of suspended and attached aggregate glue droplets are consistent (see Figs. 1c and 1f). In Fig. 1c, two spectral components and their sums are shown. The spectral shapes were well reproduced by a Voigt-fitting procedure with two peak components. The center wavelength of spectral component 1 is 500 nm and spectral component 2 is 536 nm. Hence, at least two chromophores are possessed in the aggregate glue droplet. The spectra of spectral components 1 and 2 are close to the

spectra of green fluorescent protein (GFP) and yellow fluorescent protein (YFP), respectively. The chromophore in aggregate glue droplet seems to the GFP-like proteins or the same protein family as GFP. In order to clarify the details of the non-luminous area of the attached aggregate glue droplets, it is necessary to analyze its composition.

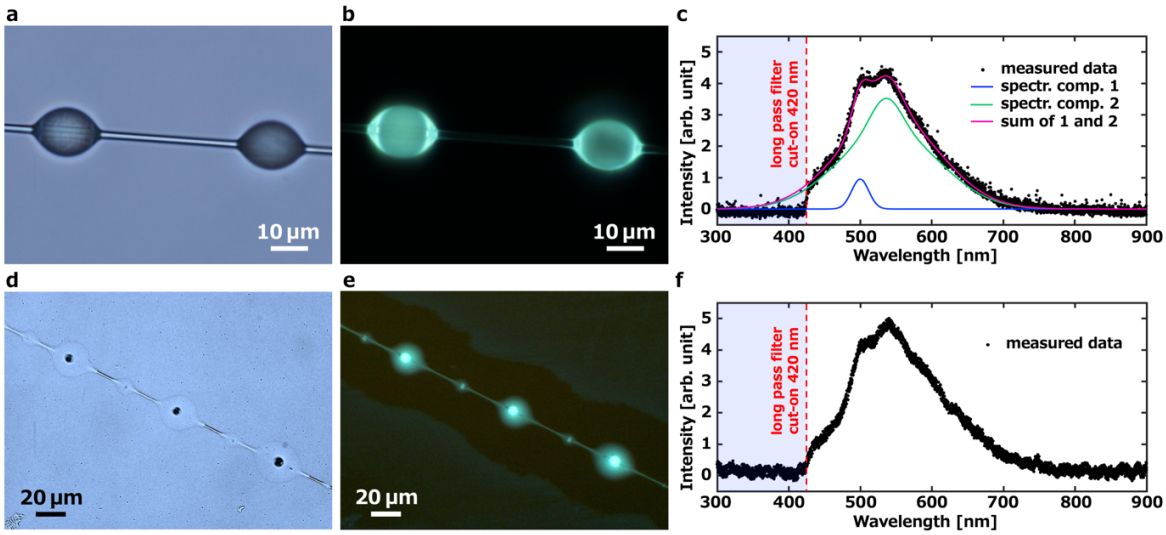

Figure 1 | Fluorescence observations of aggregate glue droplets. a, Optical microscopy image of suspended aggregate glue droplets by illuminating with white light. **b,** Fluorescence microscopy image of suspended aggregate glue droplets. **c,** Fluorescence spectrum of suspended aggregate glue droplets. The spectrum was fit with Voigt functions for spectral component 1 (blue line) and spectral component 2 (green line). The magenta line is the sums of the spectral component 1 and 2. **d,** Optical microscopy image of aggregate glue droplets attached to a glass substrate by illuminating with white light. **e,** Fluorescence microscopy image of aggregate glue droplets attached to a glass substrate. **f,** Fluorescence spectrum of aggregate glue droplets attached to a glass substrate.